A D PATEL INSTITUTE OF TECHNOLOGY
DEPARTMENT OF INFORMATION TECHNOLOGY
THE CHARUTAR VIDYA MANDAL (CVM) UNIVERSITY

## STUDENT MANAGEMENT SYSTEM

# PYTHON MINI PROJECT

**A desktop application for managing student records and academic information**

| STUDENT FORM | RECORD TABLE | SEARCH & UPDATE | LOCAL FILE STORAGE |
|---|---|---|---|

PROJECT SUBMITTED BY

**DAKSH BHAVSAR**

Enrollment No.: 12402080601022

B.Tech. (Information Technology) • Semester V

Course Code: 202044504 | PROGRAMMING WITH PYTHON | Professional Core Course

# PROJECT OVERVIEW & OBJECTIVES

## PROJECT OVERVIEW

A Python desktop application that helps an administrator add, view, search, update and delete student records through a graphical interface.

## PROBLEM ADDRESSED

Manual record keeping can be slow, repetitive and prone to mistakes. A centralized application makes student information easier to organize.

## PROJECT OBJECTIVES

- Store student details in a structured format.
- Make record searching quick and simple.
- Allow student records to be updated or deleted.
- Provide a clear, user-friendly interface.

## EXPECTED OUTCOME

- A single view of student records.
- Faster retrieval of student information.
- More consistent record management.
- Persistent storage using a local file.

# TECHNOLOGY STACK

### PYTHON

Core language for application logic, validation and file operations.

### TKINTER

Builds the desktop GUI with windows, buttons, labels, input fields and tables.

### CSV / FILE STORAGE

Stores student records locally without requiring a MySQL database.

### OBJECT-ORIENTED PROGRAMMING

Organizes the application into reusable classes and methods.

### TREEVIEW TABLE

Displays student records in a selectable, tabular format.

### EXCEPTION HANDLING

Handles invalid input and file errors with helpful messages.

*Required package: Python standard library (Tkinter and csv)*

# MAIN FEATURES

### ADD STUDENT
Enter student ID, name, course, semester and contact details.

### VIEW RECORDS
Display saved students in a structured table.

### SEARCH STUDENT
Find a record using student ID or name.

### UPDATE DETAILS
Select a student row and edit the stored information.

### DELETE STUDENT
Remove a selected record after confirmation.

### LOCAL DATA STORAGE
Save records in a CSV file so they remain available after restart.

# SYSTEM WORKFLOW

### 1 LAUNCH APP

Python starts the Tkinter window.

### 2 LOAD RECORDS

Previously saved student records are read from the local file.

### 3 ENTER DETAILS

The user fills in student information in the form.

### 4 VALIDATE INPUT

Required fields and student ID are checked.

### 5 MANAGE RECORD

Add, search, update or delete the selected student.

### 6 SAVE CHANGES

Updated records are written to storage and the table refreshes.

*Validation helps prevent missing details, duplicate IDs and invalid record operations.*

# STUDENT DETAILS & SAMPLE DATA

### STUDENT ID

Unique identifier for each student.

### FULL NAME

Student's name as entered in the form.

### COURSE

Program or department the student belongs to.

### SEMESTER

Current semester of study.

### EMAIL / CONTACT

Contact information for reference.

### STATUS

Optional field indicating active or inactive status.

*Sample records shown below are illustrative; actual values depend on user input.*

# DATA STORAGE DESIGN

### STUDENTS.CSV

Local file that stores student records in a tabular format.

### STUDENT ID

Used as a unique key for finding and updating a record.

### FIELDS

Student ID, name, course, semester and contact details.

### ADD / UPDATE

Validated records are written to the CSV file.

### DELETE

The selected student's record is removed from the stored data.

### PERSISTENCE

Data remains available between application sessions.

**Storage approach: local CSV file; no MySQL database is required.**

Records are loaded at startup and saved after successful changes.

# PYTHON PROGRAM STRUCTURE

### STUDENTMANAGEMENTAPP CLASS

Creates the Tkinter interface and coordinates student record operations.

### FORM HANDLING

Reads input fields and validates required values.

### TABLE / TREEVIEW

Shows records and supports selecting a row for editing or deletion.

### FILE FUNCTIONS

Load, save and rewrite student records in CSV format.

### VALIDATION

Checks missing fields, duplicate IDs and invalid entries.

### ERROR HANDLING

Displays clear messages for file errors or incorrect input.

*Entry point: main() creates the application window and starts the Tkinter event loop.*

# SAMPLE OUTPUT — STUDENT RECORDS

| STUDENT ID | NAME | COURSE | SEMESTER |
|---|---|---|---|
| 12401 | Aarav Patel | B.Tech IT | 5 |
| 12402 | Diya Shah | B.Tech IT | 5 |
| 12403 | Rohan Mehta | B.Tech CSE | 3 |
| 12404 | Nisha Desai | B.Tech IT | 1 |

*Illustrative sample data only; these are not real student records.*

# CONCLUSION & FUTURE SCOPE

## CONCLUSION

The Student Management System demonstrates how Python, Tkinter and local file handling can simplify student record management. It supports adding, viewing, searching, updating and deleting records.

## FUTURE ENHANCEMENTS

- Admin login and access control
- Attendance and marks management
- Export records to PDF or Excel
- Advanced filters and sorting
- Database integration if required

## KEY LEARNING

Practical experience with GUI design, file handling, validation, table selection and structured Python programming.

## THANK YOU